

BIKE PURCHASE ANALYSIS

Project Documentation

A comprehensive data analysis project exploring bicycle purchase patterns across demographics, income levels, geographic regions, and commute behaviours.

Detail	Information
Project Name	Bike Purchase Analysis
Platform	Microsoft Power BI Desktop
Dataset	RawDataSet.xlsx
Dashboard File	BIKES.pbix
Total Records	1,026 customer records
Total Fields	15 variables per record
Purchase Rate	48% (495 of 1,026 customers)
Regions Covered	North America · Europe · Pacific

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1. Project Overview

This project performs a full exploratory data analysis (EDA) on bicycle purchase behaviour across a sample of 1,026 customers. The goal is to identify which demographic, economic, and lifestyle factors are most strongly associated with the decision to purchase a bicycle. Findings are visualised in an interactive Power BI dashboard (BIKES.pbix) that enables stakeholders to filter by region, gender, and age group using built-in slicers.

1.1 Business Questions Answered

- Which customer demographics (age, gender, marital status) are most likely to purchase a bike?
- How does household income level influence purchase decisions?
- Does commute distance predict bike purchase likelihood?
- Are there significant purchase patterns across geographic regions?
- Which occupational and educational profiles correlate with bike buyers?

1.2 Key Findings at a Glance

Metric	Value	Insight
Total Customers	1,026	Full sample size
Total Purchases	495	Bikes sold
Purchase Rate	48%	Nearly half of all customers bought
Avg. Buyer Income	\$57,000	Mid-range earner profile
Avg. Buyer Age	43 years	Middle-aged adult segment
Top Region	North America	220 purchases — highest volume
Top Age Group	Senior	236 purchases
Top Income Group	Low Income	234 purchases
Top Commute Seg.	0–1 Miles	207 purchases — short-commute buyers

2. Data Source & Structure

The raw dataset is stored in **RawDataSet.xlsx** (Sheet1) and contains 1,026 rows of customer records and 15 columns. No external data sources or APIs are used.

2.1 File Inventory

File	Type	Purpose
RawDataSet.xlsx	Excel Workbook (.xlsx)	Source data — 1,026 customer records, 15 fields
BIKES.pbix	Power BI Desktop File	Interactive dashboard and all visualisations
Dashboard_ScreenShot.png	PNG Image	Static preview of the final dashboard

2.2 Data Dictionary

Each row in RawDataSet.xlsx represents one unique customer. The 15 fields are described below:

#	Field Name	Data Type	Description	Sample Values
1	ID	Integer	Unique customer identifier	25597, 12610
2	Marital Status	Categorical	Customer marital status	Single, Married
3	Gender	Categorical	Customer gender	Male, Female
4	Income	Numeric (\$)	Annual household income	\$10,000 – \$170,000
5	Income Group	Categorical	Derived income tier	Low / Medium / High
6	Children	Integer	Number of dependent children	0, 1, 2, 3...
7	Education	Categorical	Highest education level attained	Bachelors, Graduate Degree...
8	Occupation	Categorical	Primary occupation category	Clerical, Professional...
9	Home Owner	Boolean	Whether customer owns their home	Yes, No
10	Cars	Integer	Number of cars owned	0, 1, 2, 3...
11	Commute Distance	Categorical	Daily commute distance band	0-1, 1-2, 2-5, 5-10, 10+
12	Region	Categorical	Geographic region of residence	N. America, Europe, Pacific
13	Age	Integer	Customer age in years	25 – 89

1 4	Age Group	Categorical	Derived age bracket	Youth, Senior, Elderly
1 5	Purchased Bike	Categorical	Target variable — did they buy?	Yes, No

2.3 Dataset Statistics

Field	Range / Unique Values	Distribution
Marital Status	2 categories	Single: 477 (46.5%) Married: 549 (53.5%)
Gender	2 categories	Male: 525 (51.2%) Female: 501 (48.8%)
Income	\$10,000 – \$170,000	Mean: \$56,209 Buyer avg: ~\$57,000
Income Group	3 categories	Low: 493 Medium: 455 High: 78
Education	5 categories	Bachelors: 311 Partial College: 278 HS: 184
Occupation	5 categories	Professional: 280 Skilled Manual: 259 Clerical: 187
Home Owner	Yes / No	Owner: 697 (67.9%) Non-owner: 329 (32.1%)
Commute Distance	5 distance bands	0-1 Mi: 378 5-10 Mi: 197 1-2 Mi: 176
Region	3 regions	North America: 508 Europe: 316 Pacific: 202
Age	25 – 89 years	Mean: 44.1 years Buyer mean: ~43 years
Age Group	3 categories	Senior: 513 Youth: 396 Elderly: 117
Purchased Bike	Yes / No (target)	Yes: 495 (48.2%) No: 531 (51.8%)

3. Power BI Dashboard

The **BIKES.pbix** dashboard is the primary deliverable of this project. It contains a single-page interactive report with five KPI cards, six chart visuals, and a slicer panel for dynamic filtering. The green-on-black colour theme is applied consistently throughout.

3.1 KPI Cards (Top Row)

Five summary cards are displayed across the top of the report to provide immediate high-level insight:

KPI Card	Value	Description
Total Customers	1,026	All records in the dataset
Total Purchase	495	Count of customers who purchased a bike
Purchase Rate	48%	$\text{Purchases} \div \text{Total Customers}$
Avg Income Buyers	\$57K	Average annual income of bike buyers
Avg Age Buyers	43	Average age of bike buyers in years

3.2 Chart Visuals

3.2.1 Total Purchase by Gender — Donut Chart

Displays the gender split among buyers. Female customers account for 252 purchases (50.91%) and Male customers 243 (49.09%), indicating an essentially equal gender distribution among bike buyers.

3.2.2 Total Purchase by Marital Status — Donut Chart

Married customers purchased 259 bikes vs. 236 by single customers, suggesting a slight preference among married individuals — possibly reflecting dual-income households with higher disposable income.

3.2.3 Total Purchase by Age Group — Horizontal Bar Chart

Seniors lead with 236 purchases, Youth follow with 222, and the Elderly group records only 37. Seniors and Youth dominate purchase activity; the Elderly segment shows significantly lower uptake.

3.2.4 Total Purchase by Region — Vertical Bar Chart

North America leads with 220 purchases, Europe with 156, and Pacific with 119. North America's larger customer base (508 total) contributes proportionally to its higher purchase volume.

3.2.5 Total Purchase by Income Group — Horizontal Bar Chart

Low Income Earners lead with 234 purchases, Medium Income Earners follow with 220, and High Income Earners account for only 41. The result largely reflects sample-size differences between tiers.

3.2.6 Total Purchase by Commute Distance — Vertical Bar Chart

A clear inverse relationship exists: 0–1 Miles (207), 2–5 Miles (95), 1–2 Miles (83), 5–10 Miles (77), 10+ Miles (33). Short-distance commuters are the primary buyers — bikes serve as practical daily transport.

3.3 Slicer Panel

Three interactive slicers allow users to dynamically filter all visuals simultaneously:

- Region — Filter by Europe, North America, or Pacific
- Gender — Filter by Female or Male
- Age Group — Filter by Elderly, Senior, or Youth

4. Methodology & Data Processing

4.1 Data Preparation Steps

- 1. Loaded RawDataSet.xlsx into Power BI Desktop via Get Data → Excel Workbook.
- 2. Reviewed column data types in Power Query Editor and confirmed correct assignments (text for categoricals, whole number for integers, decimal for income).
- 3. Verified no null or blank values in critical fields (Purchased Bike, Region, Gender, Age Group).
- 4. Confirmed that Income Group and Age Group were pre-calculated in the source data — no additional DAX segmentation required.
- 5. Closed and applied all transformations before entering the Report view.

4.2 DAX Measures

The following calculated measures power the KPI cards in the dashboard:

Measure Name	DAX Formula	Output
Total Customers	COUNTROWS('Sheet1')	1,026
Total Purchase	COUNTROWS(FILTER('Sheet1', [Purchased Bike] = "Yes"))	495
Purchase Rate	DIVIDE([Total Purchase], [Total Customers], 0)	48%
Avg Income Buyers	CALCULATE(AVERAGE([Income]), [Purchased Bike]="Yes")	\$57K
Avg Age Buyers	CALCULATE(AVERAGE([Age]), [Purchased Bike]="Yes")	43

4.3 Calculated Columns (Source Data)

Two classification columns were pre-built in the Excel source before loading into Power BI:

Column	Segmentation Logic
Income Group	Low Income Earner: < \$40,000 Medium: \$40,000–\$100,000 High: > \$100,000
Age Group	Youth: < 35 years Senior: 35–54 years Elderly: 55+ years

5. Key Insights & Findings

5.1 Demographics

- Gender is not a significant purchase differentiator — female (50.91%) and male (49.09%) buyers are nearly equal.
- Married customers slightly outperform single customers in purchase volume (259 vs. 236), suggesting household stability or dual income supports the purchase decision.
- The Senior age group (236 purchases) slightly outperforms Youth (222), while Elderly buyers are dramatically underrepresented (37 purchases).

5.2 Income & Economic Factors

- The average bike buyer earns approximately \$57,000 per year — a mid-to-lower-middle income profile.
- Low income earners have the highest purchase count (234) primarily due to larger sample representation. After normalisation for group size, purchase rates across tiers are more comparable.
- High Income Earners (only 78 in the sample) record just 41 purchases, possibly reflecting a preference for higher-end transport alternatives.

5.3 Commute Behaviour

- 0–1 Mile commuters are the strongest buyer segment (207 purchases), confirming that bikes are primarily used as short-distance commute solutions.
- Purchase volume declines sharply beyond 2 miles. 10+ Mile commuters record the lowest purchases (33), reflecting the impracticality of cycling for very long commutes.
- This insight has strong marketing implications — short-commute urban and suburban customers should be the primary campaign target.

5.4 Regional Patterns

- North America dominates total purchases (220), consistent with its larger share of the customer sample (508 of 1,026).
- Europe records 156 purchases from 316 customers — an approximate 49% conversion rate.
- Pacific records 119 purchases from 202 customers — an approximate 59% conversion rate, the highest of any region, suggesting strong bike culture or cycling infrastructure in Pacific markets.

6. Tools & Technologies

Tool	Version / Format	Role in Project
Microsoft Excel	.xlsx	Raw data storage and preliminary grouping columns
Microsoft Power BI	.pbix (Desktop)	Data modelling, DAX measures, and visualisations
Power Query	Built into Power BI	Data import, type validation, and transformation
DAX	Power BI native	Calculated measures for all five KPI cards

6.1 System Requirements to Open Dashboard

- Power BI Desktop (free) — available at <https://powerbi.microsoft.com/desktop>
- Windows 10 / 11 (64-bit recommended)
- Minimum 4 GB RAM; 8 GB recommended for smooth slicer interaction
- Screen resolution: 1280 × 720 or higher for full dashboard visibility

7. How to Use This Project

7.1 Opening the Dashboard

1. Install Power BI Desktop if not already present (free download from Microsoft).
2. Place RawDataSet.xlsx and BIKES.pbix in the same folder.
3. Double-click BIKES.pbix to open in Power BI Desktop.
4. If prompted about the data source path, click Transform Data → Data Source Settings and update the Excel file path to match your local directory.
5. Click Refresh on the Home ribbon to reload the latest data.

7.2 Using the Slicers

The three slicers on the right side of the dashboard filter all visuals simultaneously:

- Click any slicer item (e.g., 'Europe') to filter all charts to that selection.
- Hold Ctrl and click additional items to apply multi-select filters.
- Click the eraser icon on a slicer to clear that filter and return to the full dataset.

7.3 Updating the Dataset

1. Open RawDataSet.xlsx and add or edit rows in Sheet1.
2. Ensure new rows contain values in all 15 columns.
3. Verify Income Group and Age Group values match existing categories exactly (case-sensitive).
4. Save the Excel file.
5. Return to Power BI Desktop and click Home → Refresh to pull in the updated data.

— *End of Documentation* —